

Alexander Niejadlik

Computer Engineer | Embedded Systems & Hardware Design
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EDUCATION

University of Miami — Coral Gables, FL

Aug 2022 – May 2026

B.S. Electrical Computer Engineering, Software Concentration | GPA: 3.3

Relevant Coursework: Electronics II, Embedded Systems, Computer Organization & Design, Network Client-Server, Database Design & Management, Operating Systems

TECHNICAL SKILLS

Hardware: DE1-SoC, Oscilloscope, Multimeter, Soldering, Multilayer PCB, ESP32, Arduino

Software: KiCad, PSpice, LTSpice, Quartus Prime, ModelSim, SolidWorks

Programming Languages: VHDL, Verilog, ARM Assembly, C, C++, Java, SQL

Cloud: AWS, Azure

PROJECTS

Paw Tracker (Senior Design I & II) — Team Lead & Hardware Engineer

Aug 2025 – May 2026

- Engineered a custom multi-layer PCB in KiCad integrating an ESP32 microcontroller with a NEO-6M GPS module, 3-axis accelerometer, and AD8232 analog front end for high-fidelity heart rate monitoring.
- Developed low-power C++ firmware to manage sensor data acquisition, extending battery life through deep-sleep cycles and interrupt-driven wakeups.

FPGA HPS and Linux (Embedded Microprocessors)

Aug 2025 – Dec 2025

- Configured a DE1-SoC development board to boot Linux on an ARM Cortex-A9 Hard Processor System (HPS), demonstrating hardware-software integration across the system-on-chip.
- Bridged FPGA fabric and the ARM Cortex-A9 HPS via the Lightweight HPS-to-FPGA AXI bridge, interfacing HPS peripherals like an ADXL g-sensor over I2C and controlling FPGA-side LEDs dynamically from physical switch inputs.

WORK EXPERIENCE

Engineering 3D Print Lab — Lab Assistant — Coral Gables, FL

Aug 2025 – May 2026

- Diagnose and troubleshoot hardware, electrical, and software faults across three 3D printers and their networked systems, restoring functionality with minimal downtime.
- Assist students and faculty with CAD modeling and file preparation to ensure high-quality prints.

Niejadlik Home Services — Owner & Operator — North Andover, MA

May 2021 – Present

- Founded and grew a seasonal residential service business to \$10,000 a month in revenue.
- Recruited, trained, and managed a team of up to eight employees, coordinating schedules, tasks, and projects.
- Owned all business operations including budgeting, client communications, and quality assurance.

VOLUNTEER EXPERIENCE

National Women's Shelter Network — Website Developer (Volunteer)

May 2026 – Present

- Designed a standardized, modular HTML template system for the "Our Leaders" directory, establishing a reusable components framework that streamlines future content updates for non-technical staff.
- Collaborated with executive leadership to translate high-priority outreach initiatives and resource-access requirements into functional, responsive front-end layouts.

ACTIVITIES

UMaker | IM Volleyball Captain | Surfrider | UCatholic | IM Soccer | YouTube Robotics (Hobby)